Projective Symmetry, Symmetry Anomaly, and Entanglement

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Abstract

1 Introduction

The projective symmetry can be used to classify the quantum phase (keywords: 2nd cohomology group, and Ref.[2]). On the other hand, Ref.[1] proved that the state $|\psi\rangle$ in a lattice spin system is long-ranged entangled if

$$\mathcal{T}\,|\psi\rangle = e^{iP}\,|\psi\rangle \quad , \quad \text{where} \;\; P \neq 0$$

$\mathcal{T}$ is the lattice translation operator. And which can be considered as a projective symmetry

Try to relate the *symmetry anomaly* and the *projective symmetry*, they are similar to some extend.

References

[1] Lei Gioia and Chong Wang, *Phys. Rev. X* **12**, 031007 (2022)

[2] Xie Chen, Zheng-Cheng Gu, and Xiao-Gang Wen, *Phys. Rev. B* **83**, 035107 (2011)